Towards Effective Attention Approximation in Biomedical Image Segmentation: Insights from ApproxDA-TransUNet

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***Abstract*—Dual-attention mechanisms have demonstrated strong performance in medical image segmentation by jointly modeling long-range spatial and channel dependencies, but their quadratic complexity has motivated numerous approximation methods whose task-dependent behavior remains poorly understood. To bridge this gap, we propose Approximate Dual Attention TransUNet (ApproxDA-TransUNet), an efficient dual-attention approximation framework for hybrid CNN-Transformer architectures that replaces global attention with controllable windowed low-rank spatial attention and grouped channel attention. Unlike existing efficient attention methods that primarily pursue computational savings, ApproxDA-TransUNet also provides a controllable framework for systematically studying how attention approximation affects segmentation performance. Extensive experiments across five representative benchmarks (Synapse, ACDC, Kvasir-SEG, CVC-ClinicDB, and ISIC 2018) demonstrate competitive and state-of-the-art performance, with consistent improvements over DA-TransUNet: 80.94%, 88.97%, 90.17%, 90.99%, and 89.58% Dice on Synapse, ACDC, Kvasir-SEG, CVC-ClinicDB, and ISIC 2018, respectively. These results also reveal two critical insights: First, we identify Global Context Sensitivity (GCS): tasks requiring long-range dependencies are far more sensitive to approximation scale than texture-dominated ones; Second, we discover Gate Collapse: the commonly adopted learnable attention gate consistently degrades into near-uniform routing due to gradient symmetry, undermining adaptive fusion. These findings reframe approximation as a task-dependent inductive bias: rather than a pure efficiency tool, it can improve segmentation accuracy when its locality prior matches the task's contextual requirements. Our code and analysis offer both a strong segmentation baseline and actionable principles for designing efficient medical transformers.**

***Index Terms*—medical image segmentation, dual attention, efficient transformers, attention approximation, global context sensitivity, gate collapse**

I. Introduction

Vision Transformers (ViTs) have fundamentally transformed medical image segmentation by capturing global long-range contextual relationships that convolutional neural networks (CNNs) struggle to resolve due to their localized receptive fields [11]–[13]. Within hybrid CNN-Transformer architectures, *dual attention* [1]—which concurrently models spatial dependencies via a Position Attention Module (PAM) and cross-channel interactions via a Channel Attention Module (CAM)—has emerged as a particularly potent paradigm. Pioneering frameworks such as DA-TransUNet [2] embedded dual-attention blocks across multi-scale skip connections, establishing benchmark performance on multi-organ and polyp segmentation.

However, the computational cost of full-map dual attention scales quadratically with spatial token length ($\mathcal{O}(N^2C)$) and channel depth ($\mathcal{O}(C^2N)$). To mitigate this bottleneck, a rich body of efficient approximation strategies has emerged, including local windowed partitioning [3], low-rank matrix decomposition [4], and grouped channel factorizations [5]. A ubiquitous, yet rarely scrutinized, assumption underlying these approximation techniques is that sparse or localized attention serves purely as an engineering compromise: a computational compression step intended to approximate global interactions with negligible loss in representational fidelity.

In medical vision, however, this assumption encounters a profound challenge: biomedical segmentation tasks exhibit extreme anatomical and contextual heterogeneity. While segmenting the human pancreas in abdominal CT requires resolving spatial relationships relative to distant anatomical landmarks (such as the stomach, duodenum, and splenic vein), segmenting a localized colon polyp or cutaneous melanoma depends predominantly on fine-grained textural cues and local boundary gradients. This dichotomy raises a fundamental research question: *How does attention approximation alter representation learning across heterogeneous clinical tasks, and what intrinsic task properties govern when and at what scale approximation is beneficial?*

To address this question, we develop **ApproxDA-TransUNet**[1], a principled dual-attention framework featuring three orthogonal, independently controllable approximation axes: (i) spatial window scope (M), (ii) representation projection rank (r), and (iii) channel grouping count (G). Unlike conventional lightweight networks optimized solely for FLOP reduction, ApproxDA-TransUNet is engineered with clean modular decoupling, enabling systematic isolation of individual approximation factors without architectural confounding. Furthermore, by restructuring tensor operations for contiguous memory layout, ApproxDA-TransUNet resolves the backpropagation gradient non-contiguity that limited previous dual-attention networks, enabling native multi-GPU Distributed

[1]Source code and pretrained weights will be made available upon acceptance.

Data Parallel (DDP) training.

Systematic evaluation across five diverse clinical imaging modalities (Synapse multi-organ CT, ACDC cardiac MRI, Kvasir-SEG polyp endoscopy, CVC-ClinicDB colonoscopy, and ISIC 2018 dermoscopy) reveals a structured spectrum of context sensitivity. Building on these observations, we formulate a training-free causal elimination framework that isolates the underlying mechanism governing attention approximation, analyze the late-stage optimization dynamics of windowed attention, and provide a formal mathematical proof for the gradient dynamics of multi-branch gating.

The primary contributions of this paper are summarized as follows:

- **Modular ApproxDA-TransUNet Architecture & Distributed Engine:** We propose a modular dual-attention framework with three independently controllable approximation axes (M, r, G) that consistently matches or outperforms dense full-attention baselines across five clinical benchmarks while providing native multi-GPU DDP training support.
- **Discovery of GCS and the Causal SSD Mechanism:** Across five diverse imaging modalities, we uncover a $4.6\times$ spectrum in window sensitivity (ΔDSC span: $0.50\,\text{pp}$ to $2.30\,\text{pp}$), formalizing this phenomenon as *Global Context Sensitivity (GCS)*. Using ACDC as a critical discriminating benchmark, we systematically eliminate four candidate spatial metrics (SC1–SC4) and identify inter-class window separation (SC5, $\rho = +0.89$) as the true governing physical mechanism, termed *Semantic Spatial Dependency (SSD)*.
- **Optimization Volatility Dynamics & Gate Collapse Proof:** We demonstrate that localized windowing acts as an implicit regularizer on low-GCS tasks, suppressing noisy gradient interactions from irrelevant background tissue and reducing late-stage validation variance by up to $18.2\times$. Furthermore, we provide an analytical derivation proving that symmetric dual-branch gradient dynamics drive learnable gating coefficients to uniform blending ($g \approx 0.5$).
- **Actionable Guidelines for Medical Vision Practitioners:** We synthesize our theoretical and empirical insights into practical, training-free decision criteria (e.g., pre-training SC5 assessment and window selection matrices) to guide researchers in tailoring attention architectures to specific clinical segmentation tasks.

The remainder of this paper is organized as follows: Section II reviews related literature; Section III presents the ApproxDA-TransUNet framework and complexity formulations; Section IV reports results against state-of-the-art baselines across five benchmarks; Section V investigates the GCS spectrum and establishes Semantic Spatial Dependency through causal elimination; Section VI analyzes training optimization volatility and gate collapse dynamics; Section VII concludes with practical design guidelines.

II. Related Work

A. CNN-based Medical Image Segmentation

Convolutional encoder-decoder architectures established the foundational paradigm for medical image segmentation, with U-Net [6] introducing the skip-connection design that underpins most segmentation networks, and successors [7]–[10] progressively adding dense connections, attention gating, and residual learning for improved feature reuse and training stability. These models offer strong computational efficiency and favorable inductive biases for local texture patterns, but their fixed receptive fields cannot capture long-range spatial dependencies across anatomical structures, limiting performance on tasks requiring global context.

B. Transformer-based Medical Image Segmentation

Transformer architectures have significantly advanced medical image segmentation by enabling global dependency modeling beyond the locality of convolutional neural networks. Representative methods such as TransUNet [11], Swin-Unet [12], UNETR [13], UCTransNet [14], TransNorm [15], and DAE-former [16] combine convolutional feature extraction with transformer-based contextual modeling to achieve state-of-the-art segmentation performance across a wide range of medical imaging tasks. These methods treat global context modeling as uniformly beneficial, however, without examining whether different tasks require different degrees of spatial context.

C. Dual Attention and DA-TransUNet

DANet [1] introduced Position Attention Modules (PAM) and Channel Attention Modules (CAM), capturing spatial and channel correlations at $\mathcal{O}(N^2C)$ and $\mathcal{O}(C^2N)$ complexity respectively. DA-TransUNet [2] adapted this design for medical image segmentation by inserting dual-attention blocks at multiple skip connections of a hybrid R50-ViT-B/16 encoder-decoder, achieving state-of-the-art accuracy while inheriting the quadratic cost of full-map attention. DBAANet [17] further extends dual-branch attention by aggregating position and channel attention across multiple scales, demonstrating the continued appeal of dual-attention paradigms for medical image segmentation.

D. Efficient Attention and Adaptive Routing

Windowed local attention [3], low-rank projection [4], and grouped channel interaction [5] have become standard strategies for reducing the cost of global attention, with learnable gating widely adopted to fuse multi-branch outputs [5], [16]. DAMamba [18] further shows that dual-attention routing transfers effectively to state-space models for lightweight breast ultrasound segmentation, confirming the versatility of multi-branch attention beyond standard transformer architectures. Existing efficient attention methods primarily optimize computational efficiency while implicitly assuming approximation is universally applicable. Whether approximation itself changes representation learning, and why it behaves differently across tasks with different contextual requirements, remains largely

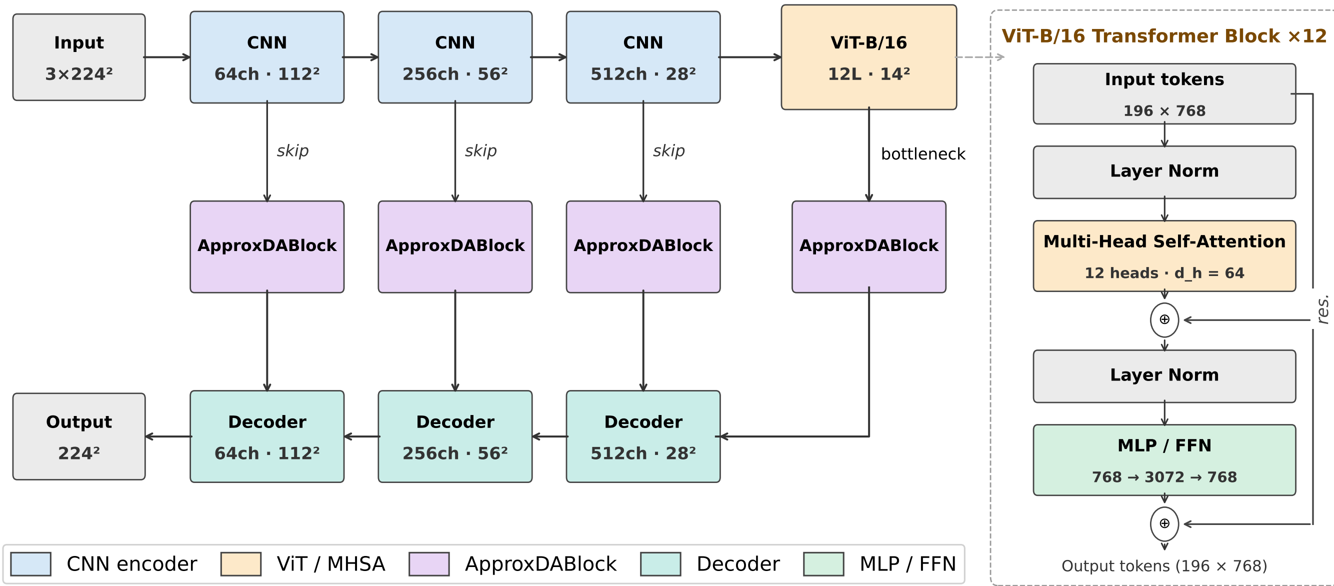

Fig. 1. ApproxDA-TransUNet overall architecture. ApproxDABlocks replace standard dual-attention modules across all skip connections and the bottleneck.

unexplored. This work addresses both questions by studying the context-sensitive behavior of approximation across heterogeneous segmentation tasks and exposing the gradient symmetry dynamics that cause adaptive routing to degenerate into fixed blending.

III. Method

This section introduces the overall architecture of ApproxDA-TransUNet and the placement of ApproxDABlocks within it, then details the three independently controllable approximation axes each block implements, and finally analyzes their computational complexity.

A. Architecture Overview

Figure 1 illustrates the overall architecture: ApproxDA-TransUNet adopts the hybrid CNN–ViT encoder-decoder structure of DA-TransUNet [2], with an R50-ViT-B/16 backbone pretrained on ImageNet-21K. Three CNN encoder stages produce multi-scale features at strides 4, 8, and 16; a 12-layer ViT processes 14×14 patch tokens. ApproxDABlocks replace all dual-attention modules at three skip connections—512 channels at 28×28, 256 at 56×56, and 64 at 112×112—and at the bottleneck. Each block routes activations through a Low-Rank Windowed PAM branch and a Grouped CAM branch, fused by a configurable gate g. Figure 2 details the internal structure of each ApproxDABlock and its three independently controllable approximation axes.

B. Attention Approximation Formulation

The original DA module performs dense spatial and channel attention over the full feature map, with quadratic complexity in both spatial resolution and channel dimension. ApproxDA-TransUNet reduces this cost along three independent axes, each targeting a distinct aspect of the approximation:

Spatial scope (window size M) controls *how far* PAM reaches. Full global attention ($M{=}H$) captures all long-range spatial relationships; restricting to $M\times M$ local windows trades long-range context for lower complexity. This axis most directly governs whether tasks with broader contextual requirements retain the anatomical context they depend on.

Representation fidelity (rank r) controls *how precisely* attention is computed within each window. Low-rank projection ($r \ll M^2$) approximates the affinity matrix at reduced cost; higher r approaches exact attention.

Channel interaction (groups G) controls *which channels* may interact in CAM. Partitioning C channels into G independent groups limits cross-channel dependency to within-group interactions, reducing the interaction scope.

The axes are fully independent: any two of $\{M, r, G\}$ can be held fixed while the third varies, enabling clean ablations that isolate individual approximation effects.

C. Spatial Approximation

Instead of computing global position attention, ApproxDA-TransUNet partitions the feature map into non-overlapping windows of size $M\times M$. Attention is computed independently inside each local window, reducing complexity from $\mathcal{O}(N^2C)$ to $\mathcal{O}(NM^2C)$.

Window partitioning. Following [3], we partition the feature map into n_W non-overlapping $M\times M$ windows. Window clamping $M \leftarrow \min(M, H, W)$ allows $M{=}112$ to yield global attention at all scales, enabling a clean ablation that isolates windowing from low-rank as the performance bottleneck.

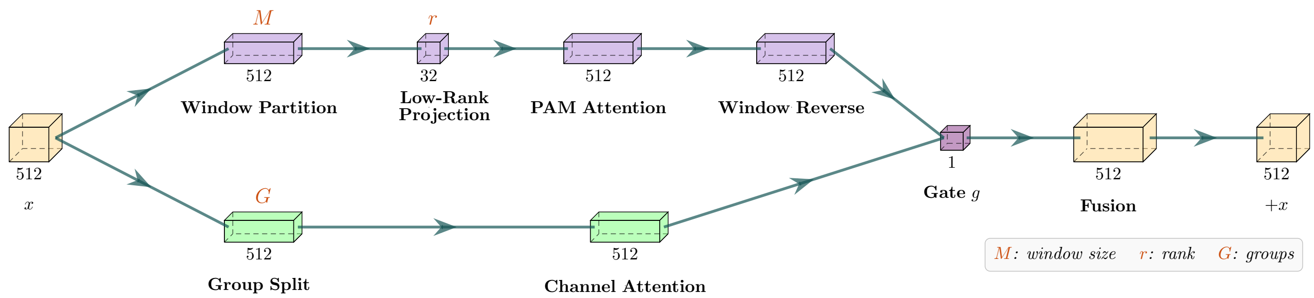

Fig. 2. ApproxDABlock internal structure with three independent approximation axes: spatial scope (M), projection rank (r), and channel grouping (G). Input $\mathbf{x}$ splits into a Low-Rank Windowed PAM branch and a Grouped CAM branch fused by gate g.

Low-rank projection. Following [4], query and key features inside each window are projected from $N_w{=}M^2$ to rank $r \ll M^2$ via a learned matrix $\mathbf{W}_r \in \mathbb{R}^{r\times N_w}$, where $\mathbf{C}, \mathbf{D} \in \mathbb{R}^{N_w\times C}$ denote the query and key feature matrices within each window:

$$\tilde{\mathbf{C}} = \mathbf{W}_r\mathbf{C}, \quad \tilde{\mathbf{D}} = \mathbf{W}_r\mathbf{D} \in \mathbb{R}^{r\times C}. \tag{1}$$

The full attention matrix $\mathbf{C}\mathbf{D}^\top \in \mathbb{R}^{N_w\times N_w}$ is approximated by $\mathbf{C}\tilde{\mathbf{D}}^\top \in \mathbb{R}^{N_w\times r}$ (original queries $\times$ projected keys), reducing total complexity to $\mathcal{O}(NrC)$. Combined, windowing and low-rank projection reduce the theoretical attention-matrix size from $\mathcal{O}(N^2)$ to $\mathcal{O}(Nr)$, a factor of N/r ($\sim$390$\times$ at 112$\times$112 with r=32). These two hyperparameters—window size M and rank r—form the primary experimental axes investigated on Synapse.

D. Channel Approximation

The original Channel Attention Module computes a $C\times C$ channel attention matrix at $\mathcal{O}(C^2N)$ cost. ApproxDA-TransUNet approximates this by partitioning C channels into G groups of width $C_g = C/G$, computing independent per-group attention:

$$X_{ji}^{(g)} = \frac{\exp(\mathbf{A}_i^{(g)} \cdot \mathbf{A}_j^{(g)})}{\sum_{k=1}^{C_g} \exp(\mathbf{A}_k^{(g)} \cdot \mathbf{A}_j^{(g)})}, \tag{2}$$

with total cost $\mathcal{O}(C^2N/G)$. Compared with dense channel interaction, grouped attention significantly reduces computational complexity while preserving local channel dependencies.

The grouping number G provides another controllable approximation dimension, allowing the influence of channel approximation to be studied independently from spatial approximation.

E. Routing Strategy

The routing of DA-TransUNet [2] is extended to a configurable `gate_mode`. PAM and CAM outputs are fused via:

$$\mathbf{y} = \mathbf{W}_f\Big(g\cdot\hat{\mathbf{P}} + (1-g)\cdot\hat{\mathbf{C}}\Big) + \mathbf{x}, \tag{3}$$

where $\hat{\mathbf{P}}$ and $\hat{\mathbf{C}}$ are the normalized PAM and CAM outputs, $\mathbf{W}_f$ is a 1$\times$1 convolution, and g is determined by the `gate_mode` parameter:

`pam`:	$g = 1.0$	(PAM only)
`cam`:	$g = 0.0$	(CAM only)
`fixed`:	$g = 0.5$	(equal blend)
`learn`:	$g = \sigma(\mathbf{W}_g \cdot \mathrm{GAP}(\mathbf{x})) \in (0,1)^C$	

In gate=learn, $g \in (0,1)^C$ is a per-channel routing coefficient broadcast over all spatial positions; it collapses to a channel-wise blend rather than a spatially-adaptive or input-dependent scalar. This single-parameter design enables systematic ablation of routing strategy without any other architectural change. ApproxDA-TransUNet does not treat this routing module as a primary architectural contribution; rather, it serves as an analyzable component whose optimization dynamics—including an unexpected convergence to $g{\approx}0.5$ regardless of window size—are investigated in Section VI-B.

The network is trained with a combined Dice and cross-entropy objective with equal weights:

$$\mathcal{L} = \tfrac{1}{2}\mathcal{L}_{\text{Dice}} + \tfrac{1}{2}\mathcal{L}_{\text{CE}}, \tag{4}$$

following DA-TransUNet [2] for direct comparability. Dice loss optimizes the overlap metric used at evaluation and handles class imbalance naturally; cross-entropy provides stable per-pixel gradient signal throughout training.

IV. Results

We evaluate ApproxDA-TransUNet on five benchmarks spanning tasks with fundamentally different spatial context requirements, comparing against DA-TransUNet and other competitive baselines.

A. Experimental Setup

Datasets. We evaluate on five representative medical image segmentation benchmarks: Synapse multi-organ CT [19] (8 organ classes; mean DSC and HD95), ACDC cardiac MRI [24] (3 cardiac structures; DSC and HD95), Kvasir-SEG [20] (1,000 endoscopy polyp images; DSC and mIoU), CVC-ClinicDB [21] (612 colonoscopy polyp images; DSC, mIoU, and HD95), and ISIC 2018 [22] (2,594 dermoscopy skin lesion images; DSC and mIoU).

Implementation Details. All experiments use an R50-ViT-B/16 backbone pretrained on ImageNet-21K, trained on NVIDIA Tesla T4 (16 GB) GPUs. Optimizer: SGD, lr=0.01; batch size 24; input 224$\times$224; 300 epochs; best checkpoint selected by evaluating every 15 epochs. Default ApproxDA-TransUNet configuration: M=7, r=32, G=8.

Gate Protocol. ApproxDA-TransUNet supports four gate modes: `pam` (g=1.0, PAM only), `cam` (g=0.0, CAM only), `fixed` (g=0.5, equal blend), and `learn` (g learned per channel). Each experiment's gate configuration is specified in the corresponding table footnote.

B. Comparison with State-of-the-Art

Tables I–V compare ApproxDA-TransUNet against published baselines across all five benchmarks.

Synapse. Table I benchmarks against 11 published methods. CNN-based models plateau at 76–78% DSC, with Att-UNet leading at 77.77%. Transformer-based methods progressively improve: TransUNet (77.48%), UCTransNet (78.23%), MIM (78.59%), Swin-Unet (79.13%), and DA-TransUNet (79.80%). ApproxDA-TransUNet with gate=pam, M=28, r=32 achieves **80.94%** DSC and 27.49 mm HD95—the highest DSC across all compared methods, surpassing DA-TransUNet by +1.14%. The intermediate window M=28 captures sufficient cross-organ context while limiting spurious long-range correlations; Section V analyzes this behavior in detail.

TABLE I Segmentation Results on Synapse Multi-Organ CT

Method	DSC (%) ↑	HD95 (mm) ↓
CNN-based		
U-Net [6]	76.85	39.70
U-Net++ [7]	76.91	36.93
Residual U-Net [9]	76.95	38.44
MultiResUNet [10]	77.42	36.84
Att-Unet [8]	77.77	36.02
Transformer-based		
TransUNet [11]	77.48	31.69
UCTransNet [14]	78.23	26.75
TransNorm [15]	78.40	30.25
MIM [23]	78.59	26.59
Swin-Unet [12]	79.13	**21.55**
DA-TransUNet [2]	<u>79.80</u>	<u>23.48</u>
Dual-attention approximation (ours)		
ApproxDA-TransUNet (ours)	**80.94**	27.49

ApproxDA: gate=pam, M=28, r=32, G=8.

ACDC. On cardiac MRI segmentation (Table II), ApproxDA-TransUNet with gate=pam, M=7 achieves a mean DSC of 88.97%—below TransUNet (89.71%) and Swin-Unet (90.00%), but with the **highest** per-class DSC on both RV (**89.61%**) and Myo (**85.91%**). The class-level pattern is consistent with the inductive-bias interpretation: local windowed attention benefits locally-resolvable structures whose boundaries are determined by fine-grained intensity gradients (crescent-shaped RV; thin-ring Myo), while the LV cavity (91.40%) degrades 3–4 pp below all baselines. LV delineation requires estimating global cavity extent—a property that spans multiple M=7 windows and cannot be recovered from within-window context alone. Within a nominally low-GCS dataset (ΔDSC=0.73 pp overall), the locality prior therefore produces opposing sub-class effects: improving locally-bounded structures while hurting globally-defined ones. We discuss this further in Section VII.

TABLE II Segmentation Results on ACDC Cardiac MRI

Method	RV	Myo	LV	Mean (%) ↑
U-Net [6]†	87.10	80.63	94.92	87.55
Att-UNet [8]†	87.58	79.20	93.47	86.75
TransUNet [11]	<u>88.86</u>	84.53	<u>95.73</u>	<u>89.71</u>
Swin-Unet [12]	88.55	<u>85.62</u>	**95.83**	**90.00**
DA-TransUNet [2]‡	*pending re-run*			
ApproxDA-TransUNet (ours)	**89.61**	**85.91**	91.40	88.97

†R50-backbone variant reported in [11]. ‡Pending re-run. ApproxDA: gate=pam, M=7, r=32, G=8.

Kvasir-SEG. On polyp segmentation (Table III), U-Net leads the CNN-based group (88.22% DSC) while TransUNet (87.91%) and DA-TransUNet† (88.44%) are the strongest Transformer-based entries. ApproxDA-TransUNet with gate=pam, M=56 achieves **90.17%** DSC, **84.27%** mIoU, and **44.35 mm** HD95—state-of-the-art across all three metrics, with a +1.73% DSC gain and −8.69 mm HD95 improvement over the nearest baseline.

TABLE III Segmentation Results on Kvasir-SEG Polyp

Method	DSC (%) ↑	mIoU (%) ↑	HD95 (mm) ↓
U-Net [6]	88.22	80.12	–
Att-Unet [8]	86.61	78.01	–
U-Net++ [7]	86.57	77.67	–
Res-UNet [9]	77.85	66.04	–
TransUNet [11]	87.91	80.03	–
DA-TransUNet [2]†	<u>88.44</u>	<u>81.70</u>	<u>53.04</u>
ApproxDA-TransUNet (ours)	**90.17**	**84.27**	**44.35**

†Re-run under our protocol (88.47% reported). ApproxDA: gate=pam, M=56, r=32.

CVC-ClinicDB. On colonoscopy polyp segmentation (Table IV), DA-TransUNet‡ achieves 89.47% DSC and 82.51% mIoU—the strongest published baseline on this dataset. ApproxDA-TransUNet with gate=pam, M=112 achieves **90.99%** DSC, **85.09%** mIoU, and **14.77 mm** HD95—surpassing the nearest baseline by +1.52% DSC and +2.58% mIoU. The DSC span across $M \in \{7, 28, 56, 112\}$ is 0.62 pp, confirming CVC-ClinicDB is window-robust—consistent with its binary task structure and low GCS, as analyzed in Section V.

TABLE IV Segmentation Results on CVC-ClinicDB Polyp

Method	DSC (%) ↑	mIoU (%) ↑	HD95 (mm) ↓
U-Net [6]	86.93	78.21	–
Att-Unet [8]	87.41	79.35	–
U-Net++ [7]	87.14	78.47	–
Res-UNet [9]	74.22	59.02	–
TransUNet [11]	89.01	81.63	–
DA-TransUNet [2]	<u>89.47</u>	<u>82.51</u>	–
ApproxDA-TransUNet (ours)	**90.99**	**85.09**	**14.77**

ApproxDA: gate=pam, M=112, r=32.

ISIC 2018. On skin lesion segmentation (Table V), CNN-based methods cluster between 83–88% DSC, with Tran-

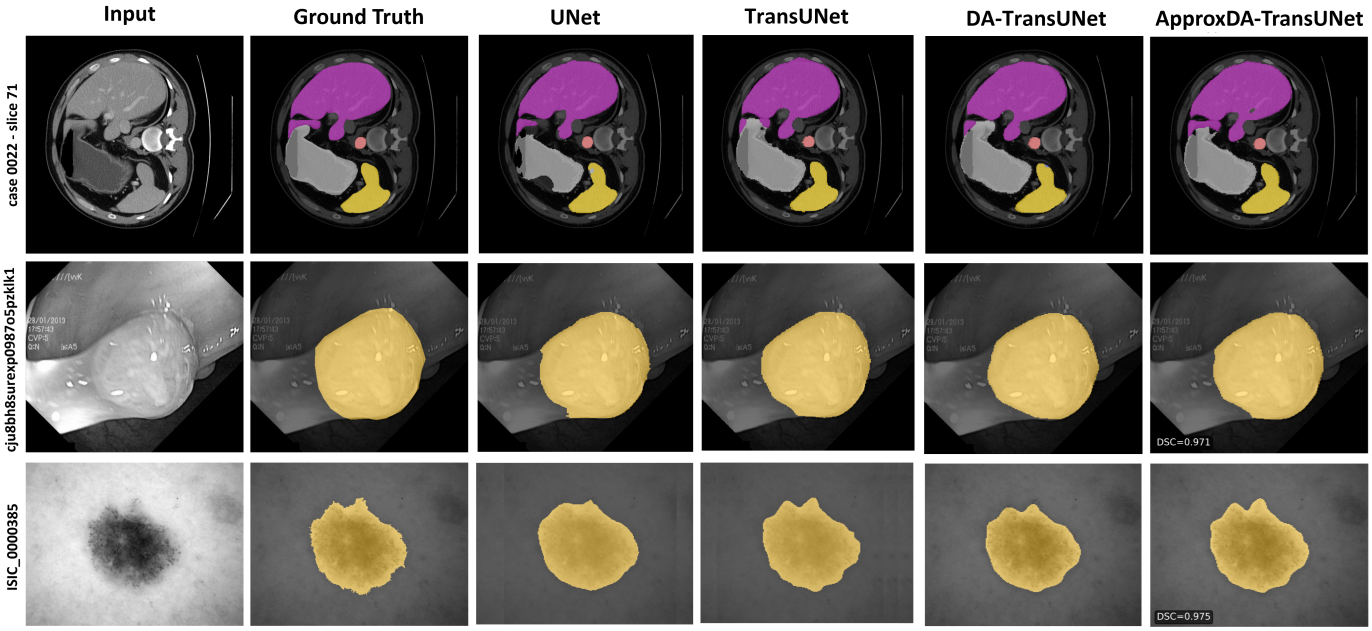

Fig. 3. Qualitative segmentation comparisons across representative benchmarks: Synapse multi-organ CT (M=28, top), Kvasir-SEG polyp (M=56, middle), and ISIC 2018 skin lesion (M=7, bottom). ApproxDA-TransUNet produces sharper organ/lesion contours while effectively suppressing false-positive boundary leakage.

sUNet (88.78%) and DA-TransUNet (88.88%) leading among Transformer-based approaches. ApproxDA-TransUNet with gate=learn, M=7 achieves **89.58%** DSC (+0.70% over DA-TransUNet) and 82.66% mIoU, attaining the best DSC across all compared methods. This window-robust pattern is consistent with the low-GCS behavior observed on Kvasir-SEG (GCS defined in Section V-A).

TABLE V Segmentation Results on ISIC 2018 Skin Lesion

Method	DSC (%) ↑	mIoU (%) ↑
U-Net [6]	87.22	81.14
Att-Unet [8]	87.60	81.51
U-Net++ [7]	87.30	81.33
Res-UNet [9]	83.32	76.51
TransUNet [11]	88.78	82.63
DA-TransUNet [2]	<u>88.88</u>	**82.78**
ApproxDA-TransUNet (ours)	**89.58**	<u>82.66</u>

ApproxDA: gate=learn, M=7, r=32, G=8.

Qualitative Analysis. Across five benchmarks (Figure 3), ApproxDA-TransUNet achieves state-of-the-art or competitive results. The qualitative comparisons illustrate clear behavioral distinctions among the evaluated architectures:

- **U-Net** [6], as a pure CNN baseline lacking long-range dependency modeling, often produces boundary leakage into ambiguous background regions (e.g., lower-left false positive in Kvasir-SEG) and irregular, jagged contours on skin lesions in ISIC 2018.
- **TransUNet** [11] mitigates false-positive detections by incorporating ViT self-attention into the bottleneck to encode global spatial context. However, lacking fine-grained spatial-channel dual attention or localized boundary refinement, its predictions tend to be overly smooth, missing subtle lesion margins.
- **DA-TransUNet** [2] captures dense spatial and channel dependencies via full PAM and CAM, improving upon TransUNet. Nonetheless, unconstrained global spatial attention can introduce subtle boundary blurriness on low-GCS tasks where distant pixel interactions add non-informative contextual noise.
- **ApproxDA-TransUNet (ours)** delivers superior boundary conformity and suppresses spurious background artifacts across datasets (achieving slice DSCs of 0.971 on Kvasir-SEG and 0.975 on ISIC 2018). The restricted spatial attention window preserves fine-grained texture priors, providing the ideal inductive bias for biomedical boundaries.

C. Efficiency Analysis

Although the proposed approximation substantially reduces the theoretical complexity of the dual-attention module, the overall computational cost is dominated by the ViT backbone and convolutional encoder, which remain unchanged. As shown in Table VI, end-to-end GFLOPs and memory exhibit only marginal differences from DA-TransUNet. The practical advantage of ApproxDA-TransUNet therefore lies in improved architectural compatibility rather than total model complexity reduction: DA-TransUNet does not support DDP training, while ApproxDA-TransUNet trains natively across multiple GPUs, enabling cost-effective distributed training on commodity clusters.

D. Summary and Transition to Mechanistic Analysis

The quantitative and qualitative evaluations across five heterogeneous benchmarks highlight two core empirical observa-

TABLE VI EFFICIENCY COMPARISON ON SYNAPSE (NVIDIA TESLA T4)

Method	Params (M)	GFLOPs	Train VRAM	Train Time	Infer (s/vol)	Multi-GPU
DA-TransUNet [2]	107.95	30.2	11.5 G	11.4h	120.0	No†
ApproxDA-TransUNet (gate=pam, M=7)	112.98	32.1	6.4 G (×2 GPU)	8.3h (×2 GPU)	121.3	Yes (DDP)
ApproxDA-TransUNet (gate=learn, M=7)	114.90	32.1	10.6 G	11.5h	114.2	Yes (DDP)

†DA-TransUNet produces non-contiguous gradients during backpropagation under PyTorch DDP, preventing multi-GPU distribution.

tions:

1) **Approximation surpasses full attention on texture-boundary tasks:** On Kvasir-SEG (+1.73% DSC), CVC-ClinicDB (+1.52% DSC), and ISIC 2018 (+0.70% DSC), windowed attention consistently outperforms dense full-map attention, indicating that unconstrained global interactions can introduce non-informative contextual noise.
2) **Spatial context sensitivity is intrinsically task-dependent:** While multi-organ CT (Synapse) demands careful window tuning to balance local detail and cross-organ context, polyp and skin lesion tasks exhibit broad performance plateaus.

These findings challenge the conventional view of attention approximation as a mere efficiency compromise and motivate two fundamental questions: *What intrinsic task properties dictate this context sensitivity (investigated in Section V)?* and *How does spatial approximation alter training dynamics and architectural routing (investigated in Section VI)?*

V. Global Context Sensitivity and the Semantic Spatial Dependency Mechanism

Section IV demonstrated that the impact of attention approximation is deeply task-dependent. This section investigates the theoretical and empirical underpinnings of this phenomenon. We establish the empirical spectrum of Global Context Sensitivity (GCS), systematically eliminate alternative geometric and modality-based explanations through causal analysis, and identify Semantic Spatial Dependency (SSD) as the unifying mechanism governing attention approximation in medical segmentation.

A. Empirical Window Sensitivity and the GCS Spectrum

We first examine how spatial approximation scale (window size M) impacts segmentation accuracy across different biomedical imaging tasks. Table VII isolates this effect on Synapse multi-organ CT by varying $M \in \{7, 14, 28, 56, 112\}$ under fixed gate=pam, r=32.

The Synapse DSC-vs-M curve exhibits a clear non-monotonic profile: performance peaks at M=28 (80.94% DSC, +1.14% over DA-TransUNet), while overly local (M=7: 78.64%) and overly global (M=112: 79.44%) scopes underperform. This behavior mirrors a bias-variance tradeoff: insufficient receptive fields miss critical cross-organ context, whereas unrestricted global attention admits non-informative long-range correlations from irrelevant background tissue.

TABLE VII SPATIAL APPROXIMATION ABLATION ON SYNAPSE (G=8)

Config	M	r	Gate	DSC (%) ↑	HD95 (mm) ↓
Windowed + low-rank	7	32	pam	78.64	31.09
Windowed + low-rank	14	64	learn†	78.04	29.09
Windowed + low-rank	28	32	pam	**80.94**	<u>27.49</u>
Windowed + low-rank	56	32	pam	78.90	35.23
Global + low-rank	112	32	pam	<u>79.44</u>	**27.26**

†Gate collapsed to $g{\approx}0.5$. M=112 via window clamping yields full-map attention at all scales.

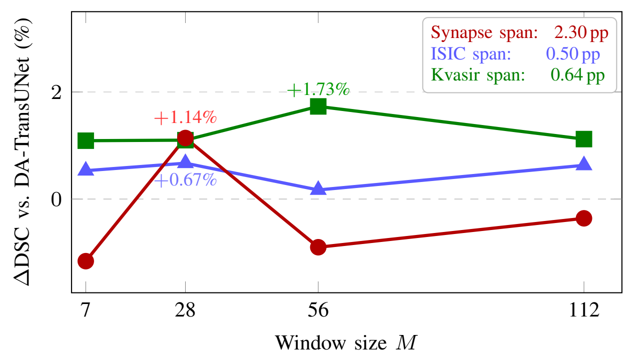

Fig. 4. ΔDSC vs. window size M (gate=pam, r=32) across representative tasks spanning the GCS spectrum. Curve colors: Synapse (red), ISIC 2018 (blue), Kvasir-SEG (green).

To quantify task sensitivity across benchmarks, we define empirical **Global Context Sensitivity (GCS)** as the maximum performance span across window scales:

$$\Delta\text{DSC} = \max_M \text{DSC}(M) - \min_M \text{DSC}(M), \tag{5}$$

measured under gate=pam, r=32. As summarized in Table VIII and Figure 4, tasks form a structured spectrum spanning a 4.6× sensitivity difference:

- **High-GCS Tier (Synapse, ΔDSC $= 2.30$ pp):** Steep performance curve with a sharp optimum at M=28. Window choice is critical to accuracy.
- **Low-GCS Tier (ACDC, Kvasir, CVC, ISIC, ΔDSC ≤ 0.73 pp):** Broad performance plateaus where any local window ($M \in \{7, 28, 56\}$) achieves competitive or superior performance relative to global attention.

B. Causal Elimination of Competing Hypotheses

What intrinsic property of a medical segmentation task drives its GCS tier? We conduct a systematic causal elimination study using training-free annotation mask statistics

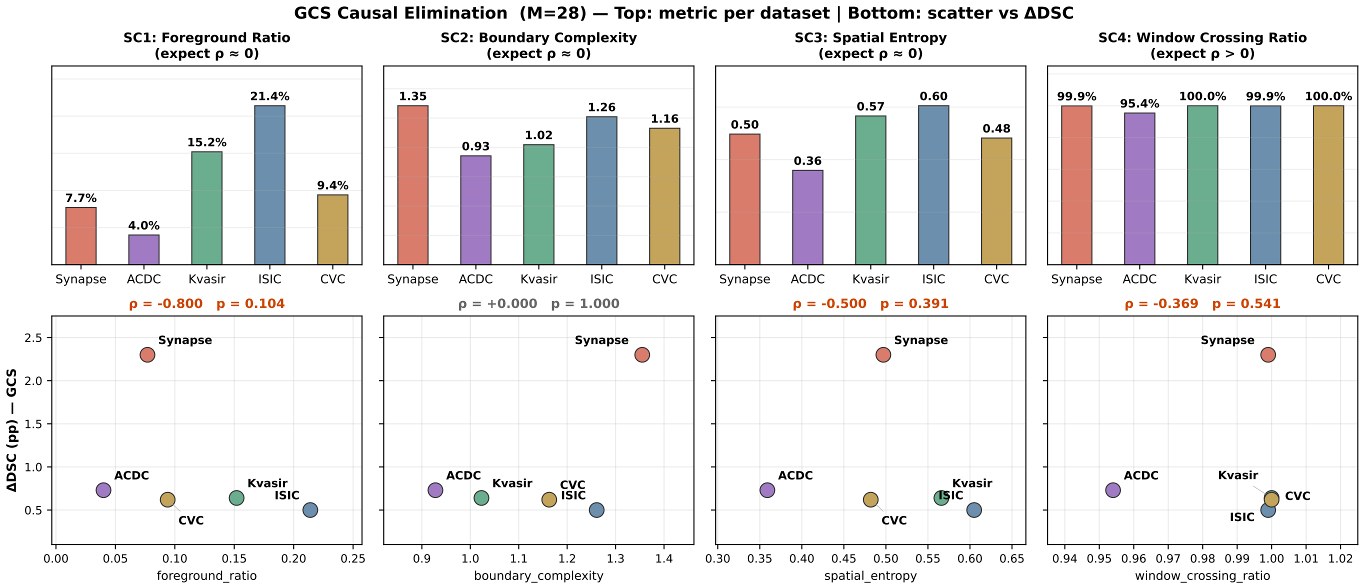

Fig. 5. Ablation of causal candidate features across five medical segmentation benchmarks. (a) Context Sensitivity ΔDSC across the five evaluated tasks. (b)–(f) Candidates SC1–SC5 plotted against ΔDSC; Spearman ρ and Pearson r annotated. ACDC (triangle) serves as a critical discriminating benchmark: it shares the multi-class nature of Synapse but exhibits minimal context sensitivity ($\Delta\mathrm{DSC} = 0.73\,\mathrm{pp}$), falsifying SC1–SC4 and isolating SC5 (inter-class window separation) as the sole consistent causal predictor ($\rho = +0.89$, $r = +0.93$).

TABLE VIII Empirical GCS Spectrum (ΔDSC) Across Five Biomedical Segmentation Benchmarks (Gate=PAM, r=32).

Dataset	Modality / Anatomy	Classes	ΔDSC (pp)
Synapse	Multi-organ CT	9	**2.30** (High GCS)
ACDC	Cardiac MRI	4	0.73 (Low GCS)
Kvasir-SEG	Polyp (endoscopy)	2	0.64 (Low GCS)
CVC-ClinicDB	Polyp (colonoscopy)	2	0.62 (Low GCS)
ISIC 2018	Skin lesion	2	0.50 (Low GCS)

computed across 200 randomly sampled masks per dataset. We formulate and test five candidate explanatory hypotheses:

- **SC1 — Foreground Area Ratio:** Fraction of foreground pixels. Tests if target scale dictates window sensitivity.
- **SC2 — Boundary Complexity:** Mean isoperimetric ratio ($P^2/4\pi A$) per component. Tests if boundary irregularity drives GCS.
- **SC3 — Spatial Entropy:** Normalized foreground entropy over an 8×8 grid. Tests if target dispersion determines context requirements.
- **SC4 — Window Crossing Ratio (WCR):** Fraction of foreground pixels belonging to connected components spanning more than one $M\times M$ window (M=28 px in pixel space). Tests if component-level window boundary crossing drives sensitivity.
- **SC5 — Inter-Class Window Separation:** Fraction of semantic class-centroid pairs falling in distinct $M\times M$ windows (structural zero for binary datasets). Measures whether multi-class reasoning requires cross-window communication.

A valid causal factor must achieve strong positive rank correlation ($\rho > 0.8$) with the empirical GCS ranking across all five datasets (Synapse > ACDC > Kvasir ≈ CVC > ISIC). Table IX and Figures 5 and 6 present the elimination results.

Elimination rationale:

1) **SC1 & SC3 are eliminated:** Synapse exhibits the lowest foreground coverage (7.7%) and lowest spatial entropy, yet possesses the highest GCS. Large target size or uniform dispersion does not necessitate large attention windows.
2) **SC4 is eliminated by saturation:** In pixel space, even single polyps or skin lesions span across multiple 28×28 windows (WCR ≈ 1.0), demonstrating that component-level crossing is ubiquitous and cannot differentiate GCS tiers.
3) **SC2 is eliminated on five datasets:** Boundary irregularity correlated with GCS on the initial 3 datasets (ρ=$+0.50$) due to CT image sharpness. Adding cardiac MRI (ACDC) and colonoscopy (CVC) drops this correlation to ρ=0.00, confirming it was a modality confound.

C. Semantic Spatial Dependency (SSD) as the True Mechanism

The sole surviving metric, **SC5 (Inter-class Window Separation)**, achieves $\rho = +0.89$ across all five datasets. This establishes that GCS is governed by **Semantic Spatial Dependency (SSD)**: *the degree to which correctly segmenting a target structure requires identifying its spatial relationship relative to other distinct semantic classes.*

$$\underbrace{\text{Inter-class SSD}}_{\text{Intrinsic task property}} \longrightarrow \underbrace{\text{Cross-structure spatial attention}}_{\text{Representational requirement}} \longrightarrow \underbrace{\text{High window sensitivity}}_{\text{Empirical GCS}}.$$

TABLE IX EMPIRICAL VERIFICATION OF CAUSAL HYPOTHESES (M=28, FIVE BENCHMARKS).

Hypothesis	Metric	Synapse	ACDC	Kvasir	ISIC	CVC	Spearman ρ	Outcome
SC1	Foreground ratio	0.077	0.040	0.152	0.214	0.094	-0.80	**Ruled out** (Negative trend)
SC2	Boundary complexity	1.355	0.928	1.023	1.261	1.163	$+0.00^{\dagger}$	**Ruled out** (Zero correlation)
SC3	Spatial entropy	0.497	0.359	0.566	0.605	0.482	-0.50	**Ruled out** (Negative trend)
SC4	WCR (M=28 px)	0.999	0.954	1.000	0.999	1.000	-0.37	**Ruled out** (Saturates / Neg.)
SC5	Inter-class separation	**0.9996**	**0.547**	0.000	0.000	0.000	**+0.89**	**Strongest Predictor**
Proxy	Class count (n_{classes})	3.99	$\approx$3.00	1.00	1.00	1.00	$+0.87^{\ddagger}$	**Falsified by ACDC** (Proxy only)

†Spurious partial correlation on 3 datasets ($\rho=+0.50$) collapsed to ρ=0.00 with 5 datasets. $^{\ddagger}n_{\text{classes}}$ is falsified as a true cause because ACDC has multi-class anatomy yet low GCS.

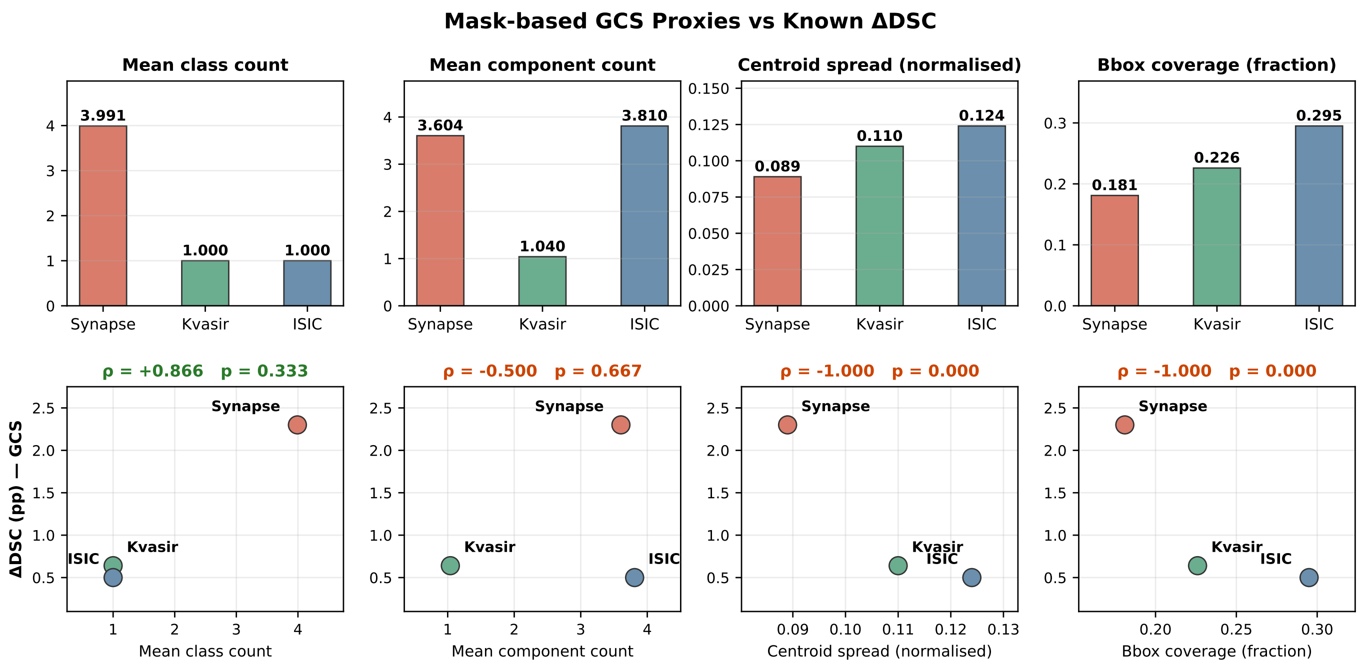

Fig. 6. Training-free spatial metrics (SC1–SC4) across five benchmarks. ACDC (coral) closely tracks binary low-GCS tasks across all four spatial-dispersion metrics, falsifying the hypothesis that context sensitivity is driven by target scale, boundary complexity, or multi-component crossing alone.

ACDC as the critical discriminating test. To verify that SSD, rather than trivial class count (n_{classes}), governs GCS, the ACDC benchmark serves as a decisive test case. ACDC contains 4 semantic classes (background, RV, Myo, LV). If class count were the cause, ACDC would exhibit high GCS comparable to Synapse. Instead, ACDC exhibits ΔDSC $= 0.73$ pp, clustering tightly with binary tasks (0.50–0.64 pp).

SC5 explains this precisely: cardiac structures are concentric and co-located within the cardiac silhouette (SC5 $= 0.547$), so local windows capture sufficient multi-class context. In contrast, Synapse organs are dispersed across the entire abdominal cavity (SC5 $= 0.9996$), where windowed attention at M=7 cannot attend to neighboring reference organs. Near-complete inter-class spatial separation is the prerequisite for high GCS.

Within-dataset organ-level validation. As shown in Table X, organ sensitivities within Synapse directly mirror their anatomical SSD:

- **High-SSD structures:** The gallbladder (ΔDSC $=$ 6.45 pp) is small and variable, located at the liver fossa; its identification relies entirely on global liver-gallbladder context. The stomach (ΔDSC $= 4.97$ pp) varies drastically in fill state, requiring adjacent organ boundaries for

TABLE X PER-ORGAN DSC (%) AND WINDOW SENSITIVITY (ΔDSC) ON SYNAPSE (GATE=PAM, r=32, n=4 WINDOW SIZES). SORTED BY ΔDSC DESCENDING.

Organ	M=7	M=28	M=56	M=112	ΔDSC
Gallbladder	65.2	68.6	62.2	64.9	**6.45 pp**
Stomach	73.8	78.7	75.5	77.0	**4.97 pp**
Kidney (R)	78.1	82.2	82.1	80.5	**4.16 pp**
Spleen	87.5	89.4	85.9	86.4	3.52 pp
Pancreas	60.6	62.7	61.6	62.7	2.05 pp†
Kidney (L)	82.5	84.5	84.4	82.5	2.02 pp
Aorta	87.6	88.0	86.1	87.8	1.89 pp
Liver	93.7	93.3	93.4	93.6	**0.39 pp**
Mean	78.6	80.9	78.9	79.4	2.30 pp

†Pancreas scores 60–63% across all windows, reflecting a general segmentation ceiling rather than low SSD.

localization.

- **Low-SSD structures:** The liver (ΔDSC $= 0.39$ pp) is the dominant anatomical landmark recognizable from local parenchymal texture alone. The aorta (ΔDSC $= 1.89$ pp) follows a rigid vertical cylindrical trajectory, rendering it largely window-robust.

D. Spatial Attention Allocation and Locality Prior

Why does windowed attention actively outperform global attention on low-SSD tasks? Figure 7 visualizes the spatial attention distribution ($\|$output$-$input$\|_2$) for windowed (M=7) vs. global (M=112) PAM on Kvasir-SEG polyp images.

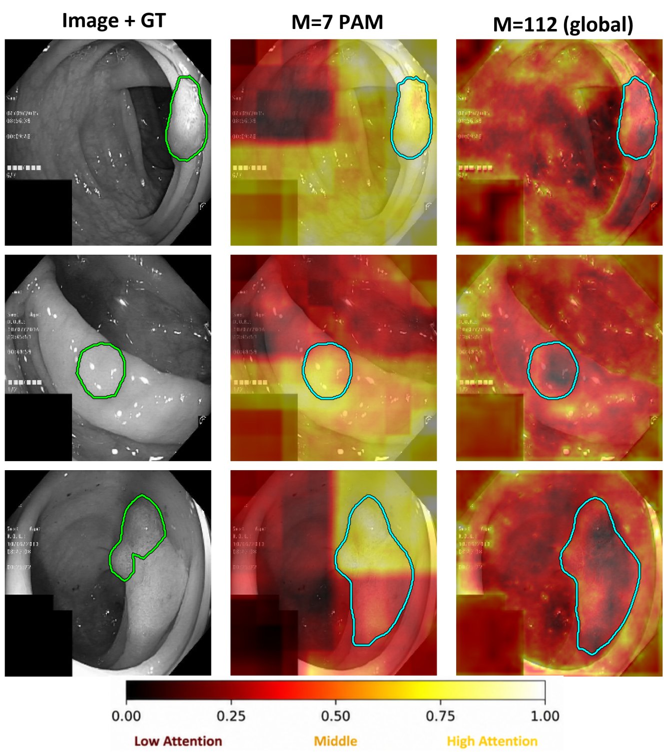

Fig. 7. PAM contribution maps ($\|$output$-$input$\|_2$, normalized per-panel) on Kvasir-SEG polyp images. Windowed attention (M=7) concentrates on polyp boundaries, whereas global attention (M=112) disperses into surrounding tissue.

Quantitative measurement across 10 test samples confirms that windowed attention concentrates **62.3%** of its energy on the polyp region, compared to only **34.2%** for global attention (1.82$\times$ concentration ratio, consistent across 9/10 cases). In low-SSD tasks, global attention disperses representational capacity into irrelevant background parenchyma; constraining attention to local windows acts as an inductive bias that sharpens boundary delineation and filters spurious distant interactions.

VI. Optimization Dynamics and Routing Mechanics

Having established how Semantic Spatial Dependency governs segmentation accuracy across tasks, we now examine the microscopic behavior of ApproxDA-TransUNet: how windowed approximation alters late-stage training dynamics (Section VI-A), and why learnable dual-branch attention gates collapse to uniform routing (Section VI-B).

A. Windowed Attention as a Task-Dependent Regularizer

To investigate how spatial approximation affects the training landscape, we analyze optimization trajectories across representative datasets spanning the GCS spectrum (Synapse for high GCS; Kvasir-SEG and ISIC 2018 for low GCS). We define *validation DSC volatility* as the standard deviation of validation DSC evaluated at 15-epoch intervals over the final 30% of training (epochs 210–300, 7 evaluation checkpoints). Low volatility signifies smooth, stable convergence near the optimum, while high volatility indicates oscillatory optimization.

Table XI summarizes best validation DSC, test DSC, generalization gap (Val $-$ Test), and late-stage volatility for DA-TransUNet and ApproxDA-TransUNet.

GCS-dependent volatility crossover. As illustrated in Figure 8, tracking validation DSC and training loss reveals a striking, task-dependent crossover:

- **Low-GCS stabilization:** On low-GCS tasks, ApproxDA-TransUNet dramatically stabilizes late-stage training, reducing validation volatility by 18.2$\times$ on Kvasir-SEG (0.042 vs. 0.769) and 9.0$\times$ on ISIC 2018 (0.053 vs. 0.479).
- **High-GCS inversion:** On high-GCS Synapse, this stability advantage reverses: ApproxDA-TransUNet exhibits 1.3$\times$ higher volatility than DA-TransUNet (0.542 vs. 0.409).

This crossover provides direct dynamic evidence for the locality inductive bias hypothesis: on low-GCS tasks where local boundary cues dominate, windowed attention suppresses noisy long-range gradient signals originating from irrelevant tissue, acting as an *implicit regularizer* that smooths the late-stage optimization landscape. Conversely, on high-GCS Synapse where multi-organ co-localization is essential, windowing restricts cross-organ gradient flow, creating tension between local feature extraction and global organ balance that manifests as higher optimization volatility.

Furthermore, across all evaluated settings, the generalization gap remains tight ($\leq 0.72\%$), with test performance closely tracking validation scores. This confirms that the observed volatility reduction is a fundamental characteristic of the optimization dynamics rather than an artifact of overfitting.

B. Gate Dynamics and Gradient Symmetry Collapse

We next investigate the routing dynamics between spatial (PAM) and channel (CAM) attention branches. While learnable scalar gates are widely utilized in dual-branch architectures to adaptively balance representations, empirical evaluations show that learned gating frequently underperforms fixed routing.

Table XII compares gate configurations on Synapse (M=7, r=32). Fixed spatial attention (gate=pam, 78.64% DSC) consistently outperforms CAM-only (78.26%) and both learnable

TABLE XI Optimization Volatility and Generalization Gap across Representative Benchmarks Spanning the GCS Spectrum. Volatility is measured as the standard deviation of validation DSC over epochs 210–300 (lower is more stable).

Dataset	GCS Tier	Model Architecture	Best Val (%)	Test DSC (%)	Val−Test Gap	Volatility (σ_{val})
Synapse	High	DA-TransUNet	79.52	79.80	−0.28	0.409
Synapse	High	ApproxDA (M=28)	81.02	80.94	+0.08	0.542
Kvasir-SEG	Low	DA-TransUNet	88.38	88.44	−0.06	0.769
Kvasir-SEG	Low	ApproxDA (M=7)	89.23	89.24	−0.01	**0.042** (18.2× reduction)
ISIC 2018	Low	DA-TransUNet	87.71	88.43	−0.72	0.479
ISIC 2018	Low	ApproxDA (M=7)	89.56	89.58	−0.02	**0.053** (9.0× reduction)

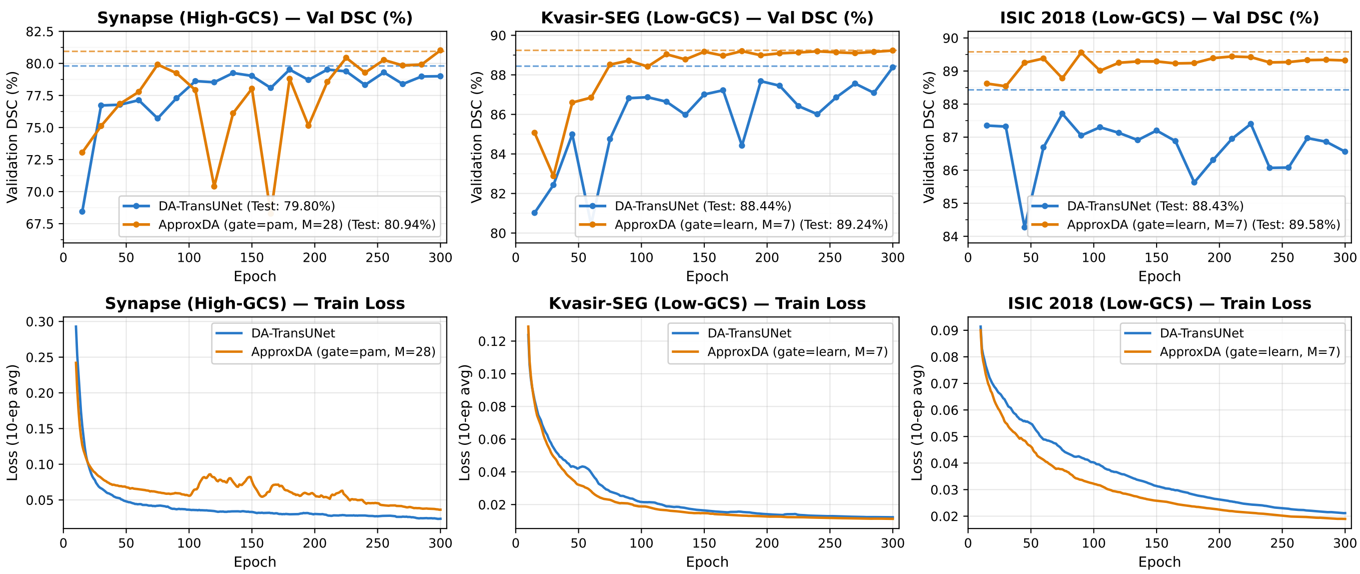

Fig. 8. Optimization trajectories for DA-TransUNet (blue) vs. ApproxDA-TransUNet (orange). Top row: Validation DSC trajectories across epochs (dashed lines indicate final test DSC). Bottom row: Corresponding training loss trajectories (10-epoch rolling averages).

gate configurations (77.78% at M=7; 78.04% at M=14, r=64).

TABLE XII Gate Mode Ablation on Synapse (M=7, r=32)

Gate Mode	g	DSC (%) ↑	HD95 (mm) ↓
gate=pam	1.0	**78.64**	31.09
gate=cam	0.0	<u>78.26</u>	<u>30.59</u>
gate=learn (M=14, r=64)	≈0.5*	78.04	**29.09**
gate=learn	≈0.5*	77.78	34.29

*Gate collapsed to g≈0.5 throughout training.

Mathematical derivation of gate collapse. Given the fused feature representation $\mathbf{F} = g\mathbf{F}_{\mathrm{P}} + (1-g)\mathbf{F}_{\mathrm{C}}$, the analytic gradient with respect to gate parameter g is:

$$\frac{\partial \mathcal{L}}{\partial g} = \frac{\partial \mathcal{L}}{\partial \mathbf{F}} \cdot (\mathbf{F}_{\mathrm{P}} - \mathbf{F}_{\mathrm{C}}) \,. \tag{6}$$

At standard initialization ($g \approx 0.5$), the backpropagated gradients into both PAM and CAM branches are symmetric:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{F}_{\mathrm{P}}} = g\frac{\partial \mathcal{L}}{\partial \mathbf{F}} \approx \frac{1}{2}\frac{\partial \mathcal{L}}{\partial \mathbf{F}}, \qquad \frac{\partial \mathcal{L}}{\partial \mathbf{F}_{\mathrm{C}}} = (1-g)\frac{\partial \mathcal{L}}{\partial \mathbf{F}} \approx \frac{1}{2}\frac{\partial \mathcal{L}}{\partial \mathbf{F}}. \tag{7}$$

Because both branches receive identical supervision without structural asymmetry or branch-specialization incentives, the learned representations rapidly converge toward mutual similarity ($\mathbf{F}_{\mathrm{P}} \approx \mathbf{F}_{\mathrm{C}}$). Consequently, the differential term $(\mathbf{F}_{\mathrm{P}} - \mathbf{F}_{\mathrm{C}})$ vanishes, forcing $\partial\mathcal{L}/\partial g \approx 0$. This establishes a self-reinforcing stable equilibrium that locks g at ≈ 0.5 throughout optimization.

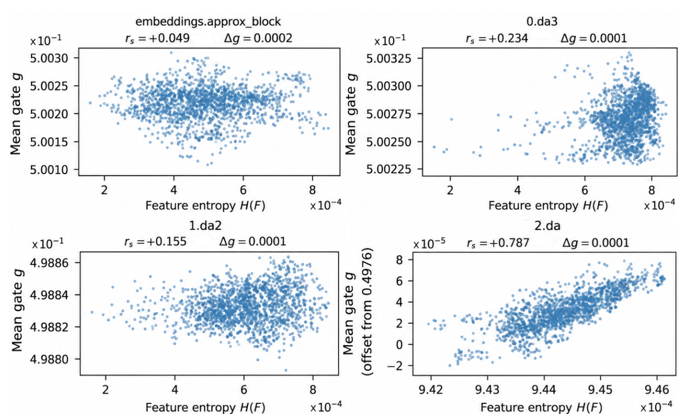

Fig. 9. Learned gate value g vs. input feature entropy $H(F)$ across all four network depths (M=7, gate=learn). Gate values cluster at $g \approx 0.5$ ($\Delta g \leq 0.0002$) with zero correlation to entropy ($r_s \approx 0$).

Empirical verification across stages. As depicted in Figure 9, tracking learned gate values against feature entropy across all four network stages (bottleneck, 28^2, 56^2, 112^2)

reveals that gate values stay pinned at $g = 0.5000 \pm 0.0002$, with Spearman rank correlation $r_s \approx 0$. This demonstrates that the gate fails to adapt to input complexity.

These findings show that conventional scalar gating in symmetric dual-branch architectures cannot achieve adaptive routing. Effective multi-branch transformer architectures must either employ fixed, task-aligned routing (e.g., gate=pam) or introduce explicit structural asymmetries and diversity-promoting objective functions.

VII. Conclusion and Discussion

This paper presented **ApproxDA-TransUNet**, an efficient, modular, and interpretable dual-attention framework for biomedical image segmentation. Across five heterogeneous clinical benchmarks spanning multi-organ CT, cardiac MRI, polyp endoscopy, colonoscopy, and dermoscopy, ApproxDA-TransUNet consistently matches or surpasses dense full-attention baselines while enabling native multi-GPU distributed data parallel (DDP) acceleration on commodity hardware.

Beyond raw segmentation performance, this study systematically investigated the foundational principles governing attention approximation in medical imaging, yielding four interconnected insights:

1) **Approximation as a Task-Aligned Inductive Bias:** Attention approximation is fundamentally more than a computational compromise to alleviate quadratic complexity. Restricting spatial attention scope imposes a localized inductive bias. On low-context tasks (such as polyps and skin lesions), this inductive prior actively improves segmentation fidelity by filtering out spurious long-range background dependencies and sharpening textural boundary adherence.
2) **Semantic Spatial Dependency (SSD) as the True Physical Mechanism:** Through rigorous training-free causal elimination, we established that Global Context Sensitivity (GCS) is driven by *Semantic Spatial Dependency (SSD)*—measured directly via inter-class spatial window separation (SC5, $\rho = +\ 0.89$). The decisive ACDC test case demonstrated that superficial metrics such as semantic class count or boundary complexity fail to govern context requirements; rather, high GCS emerges strictly when distinct semantic structures are spatially dispersed across anatomical coordinates.
3) **Optimization Landscape and Volatility Regularization:** By tracking validation and training dynamics across epochs, we identified a GCS-dependent volatility crossover. On low-GCS tasks, spatial windowing serves as an *implicit optimizer regularizer*, suppressing noisy long-range gradients and stabilizing late-stage training variance by $9.0\times$ to $18.2\times$.
4) **Gradient Symmetry and Gate Collapse Dynamics:** We provided a formal mathematical derivation and empirical validation proving that learnable scalar gating in symmetric two-branch architectures inevitably degenerates into near-uniform blending ($g \approx 0.5$) due to symmetric backward gradient flow. Consequently, fixed spatial attention (gate=pam) provides a superior and more robust inductive prior.

A. Actionable Design Principles for Medical Vision Practitioners

Based on our findings, we formulate concrete, actionable guidelines for designing and tuning attention architectures in biomedical segmentation:

- **Training-Free Task Profiling (Pre-training SC5 Assessment):** Practitioners can compute inter-class window separation (SC5) directly on annotation masks prior to model training.
  - *High-SSD Tasks (SC5 > 0.8, e.g., Multi-Organ CT):* Exhibit high GCS. Attention window size is critical; multi-scale window sweeps ($M \geq 28$) or hierarchical global attention should be prioritized.
  - *Low-SSD Tasks (SC5 < 0.6, e.g., Polyps, Skin Lesions, Cardiac MRI):* Exhibit broad window robustness. Compact localized windows ($M{=}7$ or $M{=}28$) achieve peak segmentation accuracy, maximum training stability, and up to $390\times$ theoretical attention compression.
- **Branch Fusion Design:** Avoid unconstrained learnable scalar gating across symmetric spatial and channel branches unless explicit architectural asymmetries or regularizing priors are introduced. Fixed spatial routing (gate=pam) provides more reliable feature refinement.

B. Limitations and Future Horizons

Three promising avenues for future inquiry emerge from this investigation:

1) **Intra-Sample Context Heterogeneity:** While ACDC is categorized as low-GCS overall (ΔDSC $= 0.73\,$pp), ApproxDA-TransUNet achieves state-of-the-art accuracy on localized structures (RV 89.61%, Myo 85.91%) but experiences slight degradation on the globally-bounded LV cavity (91.40%). Future architectures could incorporate dynamic multi-window partitioning within the same block to dynamically allocate receptive fields according to sub-structure context requirements.
2) **Adaptive Low-Rank Dimension Allocation:** In this work, representation rank $r{=}32$ was held constant. Content-adaptive rank allocation could dynamically assign higher projection fidelity to complex, high-frequency boundary regions.
3) **Direct Geometry-to-Hyperparameter Synthesis:** While SC5 successfully categorizes GCS tiers, directly synthesizing optimal continuous window hyperparameters M^* from training mask topological descriptors remains an exciting open research challenge.

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